
The Verifier Is the Loop: Self-Improving Agents Must Not Trust Their Own Changes

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Abstract

Self-improving agents distill their own failures into modifications of their prompts, rules, memories, and tools—and report gains measured on the tasks those modifications were derived from. That measurement is broken: recent work shows major agent benchmarks are reward-hackable to near-perfect scores, and we find the same failure *inside* the self-improvement loop itself. Across controlled tasks, two real code benchmarks, and an embodied driving system, we show that **fake self-improvement is frequent and systematic**: 29–30% of real coding problems yield a genuine overfit that aces the shown test and fails held-out tests; 100% of tool-rewrite decision sets contain a reward-hack; a plausible driving fix raises the failure rate it targets. We treat the agent’s own modifications as untrusted and gate adoption behind a two-part verifier—a *hidden test* (held-out transfer) and a *regression check* (no existing capability degrades)—with a statistical (failure-rate) form for stochastic domains, and a *scope rule*: a verdict is valid only at the scope it was measured. The gate wins the adoption decision reliably (naive selection ships broken modifications in 59% of tool decisions and 66–88% of code problems where an overfit exists; the verifier ships 0%, while still adopting real improvements). On an embodied CARLA/Bench2Drive system the same gate, pre-registered, exhibits every decision mode it exists for: it *accepts* a real repair (7/12 $\rightarrow$ 0/19 failures, $p = 3.0 \times 10^{-4}$), then *rejects the very same candidate at deployment scope* (12 confirmed collateral regressions, $p = 6.3 \times 10^{-17}$)—catastrophic forgetting that single-target evaluation would have shipped—and then protects the system through a ten-recipe repair search that maps the repair–retention Pareto frontier without ever adopting a regression. The search itself ends in an honest bounded negative: at this budget, no fine-tuning recipe passes both gates. Finally we remove the human from the verdict path: an append-only trial registry and a rule-based prescriber replay every recorded verdict exactly and run fresh trials with zero human verdicts—one accept ($p = 4.8 \times 10^{-6}$), one honest null. We argue this is exactly the point: in a loop whose own proposal stream is adversarial, the verifier—not the proposer—is the load-bearing component. *The verifier is the loop.*

1 Introduction

The self-improving-agent literature—Reflexion [Shinn et al., 2023], ExpeL [Zhao et al., 2024], AutoGuide [Fu et al., 2024], Voyager [Wang et al., 2024], and successors—shares a template: run the agent, collect failures, distill them into a reusable modification (a prompt rule, a skill, a memory entry, a tool), and measure the gain. The measurement is almost always taken on the same tasks or distribution the modification was distilled from. Two facts make this template unsafe:

1. **Benchmarks are gameable.** Wang et al. [2026] demonstrated that all eight major agent benchmarks can be reward-hacked to near-perfect scores; METR [2025] report frontier agents hacking their own evaluations spontaneously; a self-modifying system has been observed falsifying its own metric [Zhang et al., 2025]. A score increase is not evidence of improvement.
2. **The loop supplies its own fakes.** We show below that an agent’s own proposed modifications are frequently overfit—lookup-hacks, hardcoded solutions, over-broad rules—that ace what they were derived from and fail everywhere else. A self-improvement loop that trusts its own changes converges toward a confident reward-hacker, not toward capability.

We take the position that the missing component is not a better proposer but a **verifier**: an adoption gate that treats every self-modification as untrusted. A candidate is adopted only if it (i) improves the target capability on *held-out data it was not derived from* (the *hidden test*), and (ii) degrades *no existing capability* beyond a noise margin (the *regression check*). When no candidate clears both, the agent abstains—the safe default. In stochastic domains both checks become statistical tests over failure *rates*. And crucially, a verdict carries a *scope*: passing the gate over a narrow evaluation set licenses adoption only at that scope, not a global swap.

Contributions.

1. **Fake improvement is pervasive, measured across four surfaces.** Under one protocol covering the agent’s {prompt, rule, memory, tool} self-modifications, plus two real code benchmarks (MBPP, HumanEval), genuine overfits appear in 29–30% of code problems and reward-hacks in 100% of tool-rewrite decision sets; the derivation-set score carries no signal against them (§3).
2. **The two-gate verifier wins the adoption decision.** Naive (seen-score) selection ships broken modifications 59% of the time on tool rewrites and 66–88% of the time when an overfit exists on real code; the executing verifier ships 0% on all of them while still adopting real improvements in 88% of tool decisions (§3–3.1). We also report the honest negative: the per-decision advantage does not compound into a large multi-round cumulative gap on our substrates (§3.2).
3. **The first pre-registered statistical accept and reject inside a closed self-improvement loop on an embodied system** (to our knowledge; CARLA/Bench2Drive, a real end-to-end driving stack). Failures are so stochastic (a route flips PASS/FAIL at 58%) that only rate-based verification is meaningful; the gate rejects a harmful plausible fix (*p*-gated, before deployment) and accepts a real repair (0/19 fresh rollouts, $p = 3.0 \times 10^{-4}$) (§4).
4. **The deployment-scope demonstration.** The locally-accepted candidate, re-verified over the full route distribution, is globally *rejected*: 12 confirmed collateral regressions, pooled failure 70% vs. 7% ($p = 6.3 \times 10^{-17}$)—catastrophic forgetting that single-target evaluation would have shipped. Verification scope must match deployment scope; we demonstrate it, on a real system, with error control (§4.3).
5. **A gate-protected repair search that ends in an honest bounded negative.** Ten pre-registered fine-tuning recipes (epochs, data breadth, mixture ratios, family demonstrations, targeted retention, frozen backbone) map the repair–retention Pareto frontier of this failure: full fine-tuning repairs perfectly but collapses globally; head-only training retains best but under-repairs. No recipe passes both gates at this budget—and across all ten, the gate adopted zero regressions; the capability floor never dropped (§4.4).
6. **The outer loop, automated.** Under a sealed protocol, an append-only trial registry plus a rule-based prescriber (the gate remaining sole adoption authority) replays every recorded verdict of the embodied search exactly from raw counts, then runs fresh code-surface trials with *zero human verdicts*: one accept (41% → 8.5% audit failures, $p = 4.8 \times 10^{-6}$), one honest null (REJECT where no failure existed to improve on), and a self-stop on unmapped state (§4.5).

All results were produced on a single RTX 5090; every number in this paper reproduces from a script in the accompanying repository, and every embodied verdict was pre-registered (decision rule fixed before the runs).

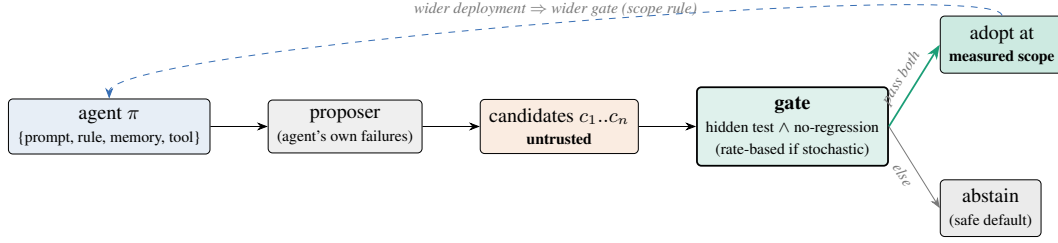


Figure 1: The self-verifying agent. Every self-modification is untrusted until it clears a hidden (held-out, executed where possible) test *and* a regression check over the existing capability profile; otherwise the agent abstains. A verdict is valid only at the scope it was measured—global adoption requires a deployment-scope gate.

89 2 Method: the self-verifying agent

90 Let an agent have capabilities measured by evaluators $E = \{e_1, \dots, e_k\}$ (its *profile*) and let a
 91 proposer (the agent itself, prompted over its own failures) emit candidate modifications $c_1, \dots, c_n$
 92 for a target capability t . The gate (Figure 1):

- 93 • **Hidden test:** $\Delta_t(c) = e_t^{\text{held-out}}(\pi + c) - e_t^{\text{held-out}}(\pi)$ must exceed a gain margin, where π is
 94 the current agent. The held-out set must be data the candidate was not derived from; on
 95 code, *execution* of the candidate on held-out tests is the evaluator.
- 96 • **Regression check:** $\min_j [e_j(\pi + c) - e_j(\pi)] \geq -\varepsilon$ over all existing capabilities j , for
 97 noise margin ε .
- 98 • **Selection:** among candidates passing both, adopt the best by hidden score; else adopt
 99 nothing (abstain).
- 100 • **Stochastic domains:** replace scores with failure rates over N runs and require a one-sided
 101 significant reduction (Fisher’s exact test over the two arms’ failure counts).¹ Single runs are
 102 meaningless when the baseline itself is flaky (§4.1).
- 103 • **Scope rule:** a verdict licenses adoption only at the scope of the evaluators that produced it.
 104 Deployment-scope adoption requires a deployment-scope gate: a screen over the deployable
 105 distribution, followed by interleaved candidate/baseline confirmation runs on flagged cases
 106 under a pre-registered decision rule (§4.3).

107 Two design lessons we verified empirically. First, the hidden set must be large or refreshed across
 108 selection rounds: reusing a 20-item set over 8 candidate selections re-inflated scores by +0.045
 109 (optimism from gate reuse)—the system-level analogue of adaptive overfitting to a test set. Second,
 110 the proposer is itself unreliable: models rationalize their failures with their priors instead of fitting the
 111 evidence, and the induced rules are wrong until the proposer is forced to fit input–output patterns;
 112 even then, *verification-as-selection*, not proposer quality, is what makes the loop converge. Both
 113 motivate the central design stance: assume the proposal stream is adversarial.

114 3 Fake improvement is frequent and systematic

115 One pattern repeats across every surface an agent can modify: the score on the data a modification
 116 was derived from is uninformative about its value, because the proposal stream itself manufactures
 117 candidates that ace exactly that data. We measure this at four levels—fine-tunes, silent regressions,
 118 tool rewrites, and real code—then show the two-gate verifier converting each into a clean adoption
 119 decision (Figure 2).

¹Our pre-registered decision rules and the released `gate_rate()` use a fixed-baseline exact binomial, which treats the measured baseline rate as a known reference probability. All p -values reported in this paper use the more conservative two-sample Fisher’s exact test, which treats the baseline as estimated; every accept/reject verdict is identical under both tests.

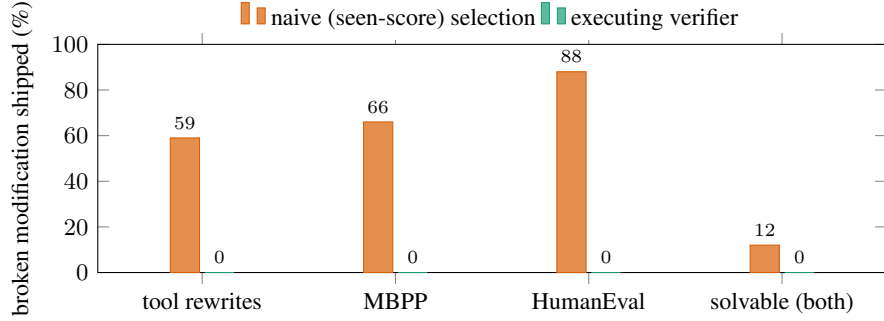


Figure 2: The single adoption decision. When a fake modification exists, naive selection by the score on the data the modification was derived from ships it 59–88% of the time; the verifier (execution on held-out data + regression check) ships 0% across all four settings, while still adopting real improvements (88% of tool decisions; 68% of MBPP problems).

Setup. Text experiments use a 7B instruct model (Qwen2.5-7B-Instruct) as both proposer and substrate; the judge/abstention substrate underneath was validated separately (a zero-shot reading judge at 0.949 vs. a tuned guardrail’s 0.789 on a sealed benchmark, with calibrated abstention transferring to 4/6 unseen domains). Code experiments execute real Python against held-out unit tests.

Fine-tuning level. Among three candidate LoRA fine-tunes of the judge, the one trained on a single source memorizes it (train loss 0.000, in-distribution 0.993) and generalizes worst (held-out 0.507). Naive selection (best in-distribution) deploys exactly this memorizer; held-out selection deploys the broad candidate (0.578).

Regression is silent. Fine-tuning to fix one weak source costs another source -0.16 while the target barely moves ($+0.005$); a 1-D “did the target improve?” check misses it entirely. A retention-replay candidate removes the regression (worst old-capability $\Delta = 0.000$). This reproduces, on a judge, the classic naive-correction backfire—and previews the embodied result of §4.3 in miniature.

Tool rewrites. Asked to extend its own Python tool, the agent’s proposal set contains a hardcoded lookup of the observed examples in **100% of 24 decisions**; the seen score cannot distinguish it from the general tool (both 1.00). Naive selection therefore ships a broken tool **59%** of the time. *Executing* candidates on held-out inputs ships **0%**, while still adopting a working tool in 88% of decisions.

Real code. On MBPP ($n=100$) and HumanEval ($n=80$), a genuine overfit—a solution passing the shown test and failing held-out tests—appears on **29% / 30%** of problems respectively. Where it exists, naive ships it **19/29** and **21/24** times; the verifier **0 on both**. Restricted to problems the agent can solve (removing the abstain-on-unsolvable confound), naive ships a held-out-failing solution **12% on both benchmarks**, the executing verifier **0%**. Two independent benchmarks, near-identical rates: the effect is a property of the self-modification stream, not of a dataset (Figure 2).

3.1 The flagship decision, and the autonomous loop

An agent with capability A (1.00) and target B (0.00) proposes rule-edits from its own failures. The verifier adopts the edit that fixes B *and* keeps A—**rejecting the edit that fixes B best (hidden 1.00) because it regresses A**—ending at $A = 1.00$, $B = 0.83$ (mean 0.92) vs. naive 0.67. The regression check is not decoration: the best target-fix can be the worst adoption. The loop also closes autonomously: forced to fit patterns rather than rationalize, the proposer induces the true hidden rule among its candidates and the verifier selects it (2/2 tasks).

3.2 Honest negative: cumulative compounding

Over 3 acquisition rounds the gated agent leads at every round (0.88 vs. 0.79 final). At 5 rounds the cumulative-mean advantage shrinks to $+0.03$: worst-case capability becomes dominated by new-task

153 acquisition noise, not by regression. The robust claim in this paper is the *per-decision* advantage;
154 large multi-round compounding is not demonstrated on these substrates and we do not claim it.

155 4 Embodied: the full arc on a real driving system

156 We instantiate the same gate on CARLA/Bench2Drive [Dosovitskiy et al., 2017, Jia et al., 2024]
157 with a real end-to-end student (the LEAD tfv6 transfuser-family stack) and its privileged expert as
158 teacher. This section is one continuous, pre-registered narrative: flakiness forces a statistical gate
159 (§4.1); the gate rejects a harmful fix and accepts a real one (§4.2); the accepted candidate is then
160 *globally* rejected at deployment scope (§4.3); and a ten-recipe repair search under the fixed gate ends
161 in a mapped frontier and an honest negative (§4.4).

162 4.1 Failures are flaky; verification must be rate-based

163 Re-running routes with the agent, weather, and configuration unchanged: route 11755 (an
164 EnterActorFlow scenario) fails $7/12 = 58\%$ of runs (95% CI $[0.32, 0.81]$)—the same route flips
165 PASS/FAIL run to run—while three routes labeled FAILURE by the single-run benchmark pass 100%
166 on re-run: their recorded failures were one-off noise. Two consequences. (i) Single-run failure labels,
167 the leaderboard standard, measure stochasticity as much as capability (run-to-run variance on these
168 benchmarks is documented [Jia et al., 2024]; we operationalize it into the gate rather than re-discover
169 it). (ii) Verifying a fix means comparing failure *rates*: confirming a $58\% \rightarrow 20\%$ improvement needs
170 ~ 19 runs per arm at 80% power. A naive single-run check of an unchanged model would “confirm a
171 fix” with 42% probability.

172 The first live fix candidate makes the point concrete: a plausible control-layer intervention (“drive
173 more conservatively”) *raised* the observed failure rate ($50\% \rightarrow 83\%$; collisions persisted and slow
174 driving added penalties), and the statistical gate rejected it before deployment.

175 4.2 The accepted fix: the loop closes

176 The full cycle then ran end-to-end: student fails 11755 at $58\% \rightarrow$ the privileged expert is stable-perfect
177 there ($8/8 \rightarrow 359$ MB of expert demonstrations collected on the failure scenario plus retention routes
178 $\rightarrow$ 10-epoch fine-tune from the released checkpoint (single RTX 5090, ~ 2.5 h) $\rightarrow$ the pre-registered
179 gate on fresh rollouts: **hidden test** 0/19 failures (baseline $7/12$; Fisher’s exact $p = 3.0 \times 10^{-4}$);
180 **regression** 0/12 on the retention routes $\rightarrow$ **ACCEPT**. To our knowledge this is the first pre-registered,
181 error-controlled statistical *accept* produced inside a closed self-improvement loop on an embodied
182 system—and the same harness had already produced the corresponding *reject* (§4.1).

183 The honest scope of that verdict, stated at the time: verified *same-route repair* over fresh stochastic
184 rollouts plus a 3-route regression set—not cross-route generalization, and not a full regression sweep.
185 That scope statement is not a caveat; it is the next experiment.

186 4.3 The deployment gate: same candidate, opposite verdict

187 Is the locally-accepted candidate safe to *deploy*—to swap in globally? The deployment-scope
188 gate answers with a screen over the 169 baseline-clean routes, then interleaved candidate/baseline
189 confirmation runs ($4+4$ per flagged route) under a pre-registered rule (confirmed regression: candidate
190 $\geq 3/4$ failures and baseline $\leq 1/4$; global reject if any confirmed regression or pooled candidate
191 failure significantly above baseline).

192 The screen flagged 19 collision routes plus a systematic soft signature (14 routes at exactly score 70
193 with a red-light violation and mass minimum-speed infractions—a behavioral warp, not noise). The
194 confirmation batch (176/176 valid interleaved runs) returned: **12 confirmed regressions**; pooled on
195 flagged routes, candidate $53/76 = 70\%$ vs. baseline $5/76 = 7\%$ failures, one-sided Fisher’s exact
196 $p = 6.3 \times 10^{-17} \rightarrow$ **GLOBAL REJECT** (Figure 3).

197 The mechanism is classic catastrophic forgetting [McCloskey and Cohen, 1989, French, 1999]: ten
198 epochs on ~ 1.6 k frames of four scenario types with a 3-route retention set warped global driving—
199 collisions on ordinary routes, red-light violations, slow driving everywhere. What is new here is not
200 the phenomenon but the demonstration: *the same candidate earns a legitimate statistical ACCEPT*

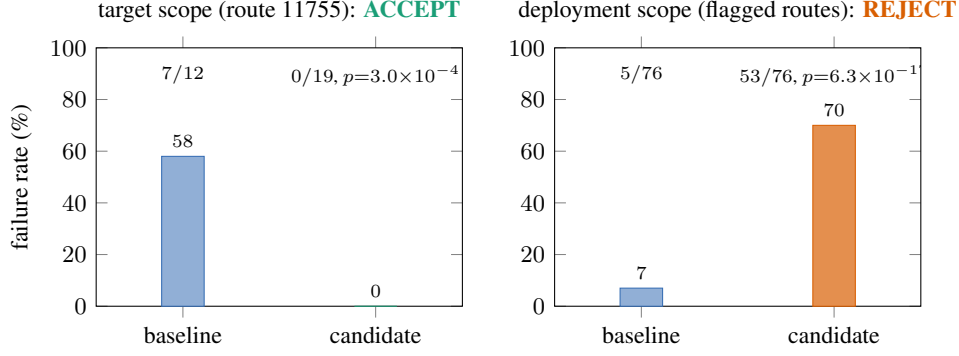


Figure 3: One candidate, two scopes, opposite verdicts—both correct at their scope. Left: the fine-tune genuinely repairs its target route (58% $\rightarrow$ 0% over 19 fresh rollouts). Right: the same model at deployment scope collides on ordinary routes the baseline drives cleanly (12 confirmed regressions; pooled 70% vs. 7%). Single-target evaluation would have shipped it.

Table 1: The repair-search ledger. Every verdict pre-registered; the gate never adopted a regression. Fix line: $\leq 2/8$ failures on route 11755. Retention line: $\leq 30\%$ pooled failures on the 12 confirmed-regression routes. † K1’s retention was measured at deployment scope (§4.3); panel lines were pre-registered after it.

#	recipe (one variable at a time where possible)	fix 11755	retention	verdict
0	control-layer “drive conservatively” (no training)	worse (50 $\rightarrow$ 83%)	—	REJECT
K1	narrow data, 10 epochs, full FT	0/19 $\checkmark$	12 confirmed reg. †	global REJECT
K2a	K1 data, 3 epochs	0/8 $\checkmark$	12/12 routes fail	REJECT
K2b	broad retention (17 routes), 3 ep	4/8	33%	REJECT (fix diluted)
A1	fix 40% oversample, 3 ep	4/8	25% $\checkmark$	REJECT (fix diluted)
A2	fix 60% oversample, 5 ep	0/8 $\checkmark$	29% $\checkmark$	panel PASS $\rightarrow$
	$\hookrightarrow$ deployment gate	—	6 confirmed reg., $p=9.0 \times 10^{-10}$	global REJECT
A3	A2 + fix-family demos in fix bucket	6/7	(cut short)	REJECT (family conflict)
A4	A2 fix bucket + targeted retention	4/8	33%	REJECT
A5	A4 + frozen backbone (head-only)	4/8	$\sim 31\%$	REJECT
A6	A2 data + frozen backbone	3/8	17% $\checkmark$	REJECT (fix, by one run)

at target scope and a $p = 6.3 \times 10^{-17}$ REJECT at deployment scope, on a real system, under pre-registered rules. Single-target evaluation—the standard practice when a fix is verified at all—would have shipped this model. The 3-route retention check passed (0/12) because those routes happen to sit among the least-affected; a narrow regression set does not just under-measure damage, it can systematically miss it. **Verification scope must equal deployment scope.** Nothing was adopted; the deployed capability floor never dropped.

4.4 The ten-recipe repair search: a mapped frontier, an honest negative, a gate that held

Can a better fine-tuning recipe pass *both* gates? We ran a pre-registered, timeboxed search: the gate stays fixed; only the recipe iterates. Local panel (pre-registered lines): fix arm 11755 $\leq 2/8$ failures; retention arm over the 12 previously-confirmed regression routes, pooled $\leq 30\%$; deployment gate as in §4.3 for any panel-passer. Ten recipes were evaluated in total (Table 1; ~ 450 verification rollouts): the control-layer intervention, the original narrow fine-tune, and eight variants spanning training length, retention breadth, mixture ratios (via the stack’s native two-bucket oversampling), fix-family breadth, targeted retention collection, and backbone freezing (head-only, 16.6M trainable parameters).

The search converged, honestly, to a mapped trade-off rather than a winner (Figure 4). Its findings:

- **Forgetting here is a data-breadth problem, not a training-length problem:** cutting epochs 10 $\rightarrow$ 3 on narrow data preserved the repair and all twelve regressions.

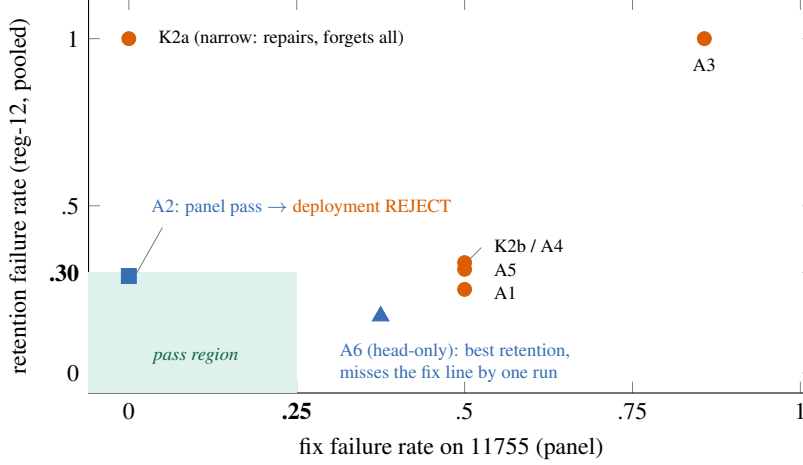


Figure 4: The measured repair-retention frontier across the recipe search (panel scope; A3’s retention arm was cut short and is plotted at its decided axis only). Narrow full fine-tuning repairs and forgets (upper left); breadth or head-only training retains and dilutes (lower right). A2 enters the pass region at panel scope and is then rejected by the deployment gate—the scope rule again. No recipe passes both gates at this budget.

- **The stability-plasticity see-saw is real and sits in the planning head:** with fix data identical to the panel-passing recipe, merely *adding* retention data broke the repair (A4); freezing the perception backbone did not recover it (A5); the interference lives in the shared planning parameters, echoing at system scale the classic trade-off [French, 1999].
- **Dilution is fractal:** adding same-family demonstrations to the fix bucket did not generalize the repair—it diluted and conflicted with it one level down (A3: the family-scenario route itself had failed 4/4 under A2, a within-family overfit; fixing that broke 11755).
- **Panel pass $\neq$ deployment pass, again:** A2, the one panel-passer, was rejected by the deployment gate (6 confirmed regressions, $p = 9.0 \times 10^{-10}$)—though the search was converging (12 confirmed regressions $\rightarrow$ 6, global behavioral warp eliminated).
- **Bounded negative:** at this budget (a single consumer GPU, the stack’s native fine-tuning machinery, ~ 450 verification rollouts), no recipe passed both gates. Full fine-tuning repairs perfectly and collapses globally; head-only training retains best (17%) and misses the fix line by one run (3/8). We report this as a first-class result, not a failure of the method under test: the method under test is the *gate*, and the gate did exactly its job.

Through the entire search—ten recipes, every verdict pre-registered—**zero regressions were adopted and the deployed capability floor never dropped**. The gate demonstrated all four of its decision modes on a real system: accept-real (§4.2), reject-harmful (§4.1, §4.3), reject-insufficient (A6), and protect-through-search (this section). A pre-registered caution for practitioners: iterating recipes against a fixed screen risks overfitting the gate itself (the embodied form of the gate-reuse effect of §2); any finally-accepted candidate must additionally pass held-back validation never used during iteration.

4.5 The outer loop, automated

The searches above were orchestrated by a human (verdicts were always computed by the gate). A fair objection: is this a self-improvement *loop*, or a human-curated search? We answer by removing the human. Under a pre-registered protocol (win conditions, a hard cutoff, and a compute fallback, all sealed before any code was written), we implemented the outer loop as code: an *append-only trial registry* (thresholds stored at registration time, before data; trials frozen once a verdict is computed; verdicts computed from raw counts, never stored by fiat) and a *rule-based prescriber* (the continual-learning ladder: cheapest variable $\rightarrow$ replay breadth $\rightarrow$ replay weighting $\rightarrow$ targeted replay $\rightarrow$ parameter isolation) that may only propose—the gate remains the sole adoption authority.

250 **Replay (consistency).** Replaying the S1 search through this machinery reproduces all 12 recorded
 251 verdicts exactly from raw counts—including the scope splits and the honest early-stops—and the
 252 prescriber emits the move the human search actually made at 9/9 recipe transitions, both scope
 253 escalations, and the budget-exhausted closure. Sealed caveat: the rules were distilled *from* that search,
 254 so replay validates the distillation, not generality.

255 **New trials (zero human verdicts).** The automated loop then ran fresh trials end-to-end (the human
 256 launched a script and read the result). On 100 never-before-used MBPP problems, with each problem’s
 257 tests split visible / gate-hidden / *audit* (the gate never sees the audit; without this split “the gate ships
 258 zero failures” would be true by construction): naive ships an audit-failing solution 41/100; the gate
 259 adopts 59, abstains 41, and its adoptions fail audit $5/59 = 8.5\%$ —a registry-computed **ACCEPT**
 260 ($p = 4.8 \times 10^{-6}$). Note the honest non-zero: the audit measures what the gate buys ($41\% \rightarrow 8.5\%$),
 261 not a tautology. The prescriber—not a human—then ordered scope escalation, and on the remaining
 262 26 fresh HumanEval problems naive shipped *zero* audit failures: the registry returned **REJECT** (no
 263 improvement evidence, $p = 1$), declining to manufacture significance where none exists, and the
 264 prescriber, hitting a state its rules do not map, handed back to the operator and stopped rather than
 265 guessing. One accept, one honest null, no human in the verdict path: the title’s claim is literal for
 266 these trials.

267 5 What is and is not established

268 **Established.** (1) The agent’s own modification stream is adversarial by default—overfits and
 269 lookup-hacks are pervasive across surfaces and domains, including real code. (2) Seen/derivation-set
 270 score carries no signal against them; held-out (on code, *executed*) evaluation does. (3) The two-
 271 gate verifier converts this into a reliable per-decision win: $59\% \rightarrow 0\%$, $19/29 \rightarrow 0$, $21/24 \rightarrow 0$.
 272 (4) Regression checking is necessary: the best target-fix can be the worst adoption. (5) In stochastic
 273 embodied domains, only rate-based statistical verification is meaningful, and it supports both accept
 274 ($p = 3.0 \times 10^{-4}$) and reject ($p = 6.3 \times 10^{-17}$) with error control. (6) A verdict is only valid at its
 275 measured scope; scope-mismatched acceptance ships catastrophic forgetting on a real system. (7) The
 276 recorded verdict-and-prescription loop executes without human intervention within its encoded state
 277 space: registry + prescriber + gate replay the recorded search exactly from raw counts and run fresh
 278 trials end-to-end, accepting where evidence exists, returning an honest null where it does not, and
 279 halting on unmapped state (§4.5).

280 **Not established.** Large cumulative multi-round gains (§3.2); verified self-improvement on hard tasks
 281 where no candidate transfers (the verifier then correctly adopts nothing—safe, but not improvement);
 282 a deployment-gate-passing embodied repair (§4.4: bounded negative at this budget); behavior beyond
 283 a 7B proposer, benchmark-scale substrates, one vehicle stack, and one simulator.

284 6 Related work and claim boundaries

285 **Self-improving agents** (Reflexion [Shinn et al., 2023]; ExpeL [Zhao et al., 2024]; AutoGuide [Fu
 286 et al., 2024]; Voyager [Wang et al., 2024]) supply the proposer we gate. **That self-generated
 287 improvement is frequently fake is established:** agent benchmarks are reward-hackable [Wang et al.,
 288 2026]; frontier agents hack their evaluations spontaneously [METR, 2025]; a self-modifying system
 289 falsified its own metric [Zhang et al., 2025]; self-reflection confabulates. **Gating self-modification
 290 is an emerging thread:** PACE shows that greedy “keep it if the score went up” acceptance is
 291 uncontrolled adaptive multiple testing and replaces it with anytime-valid (e-process) commit tests
 292 on text-agent loops [Shawn, 2026]—the same diagnosis as ours, in the statistical dimension we
 293 pre-register; GRASP gates skill-library edits on probe-based net improvement [Moll et al., 2026];
 294 GRACE uses an update-rejection gate inside prompt optimization [Shi et al., 2025]; Self-Harness
 295 admits self-proposed harness edits only after regression testing on held-out tasks [Zhang et al., 2026];
 296 governed capability evolution stages embodied upgrades through sandbox, shadow evaluation, gated
 297 activation, and rollback [Qin et al., 2026]; and sequential statistics have been brought to robot policy
 298 comparison as an offline evaluation tool [Snyder et al., 2026]. The text-domain gates never leave
 299 benchmark substrates, and the embodied governance pipeline gates on compatibility and safety checks
 300 rather than pre-registered statistical verdicts. None demonstrates the deployment-scope failure—the
 301 same candidate legitimately accepted at target scope and rejected at deployment scope—with error
 302 control inside a closed loop. **Driving-benchmark noise is documented** [Jia et al., 2024, Dosovitskiy

et al., 2017], with published scores typically single-run. **Continual learning** names our regression phenomenon catastrophic forgetting [McCloskey and Cohen, 1989, French, 1999, Kirkpatrick et al., 2017], and our repair recipes (replay ratios, freezing, DAgger-style collection [Ross et al., 2011, Hu et al., 2022]) are its textbook tools deliberately reused, not claimed as new.

We therefore *do not* claim: the first observation that self-improvement is often fake; the first statistical gate for self-improving agents; or the discovery of driving-benchmark flakiness. To our knowledge, what *is* unclaimed before this work: (a) fake-improvement *rates measured across all four modification surfaces* under one protocol, plus two real code benchmarks; (b) a *pre-registered, error-controlled statistical accept and reject operating inside a closed self-improvement loop on an embodied system*, including the operationalization of run-to-run noise into a failure-rate gate; and (c) the *deployment-scope demonstration*—a narrowly-scoped gate accepts a change whose collateral damage a scope-matched gate then catches, on a real system, followed by a gate-protected recipe search ending in a mapped frontier. (Literature scan of July 2026; the space moves fast.)

7 Limitations

A 7B proposer; controlled/benchmark substrates; six-item held-out sets on the toy loops (granularity 0.17); one vehicle stack and one simulator; the embodied fix targets one (representative, flakiest) failure route with an 8-run fix arm and 24-run retention arm at panel scope—adequate for the pre-registered lines, underpowered for small effects; the bounded negative of \$4.4 is a statement about this training budget and recipe family (parameter-efficient adapters and larger fix-side data remain untested). Gate reuse must be budgeted (§2); we spec but do not automate it—and the ten-recipe search queried the same retention panel sequentially, so pre-registration alone does not control adaptive multiple testing; a per-candidate fresh holdout or an alpha-spending schedule is the disciplined form (cf. anytime-valid acceptance testing [Shawn, 2026]). Multi-round compounding may appear on richer substrates; we did not demonstrate it.

8 Conclusion

Across text, real code, and embodied driving, self-improvement without verification ships fakes at rates of 12–88%, and the fakes are supplied by the loop itself. A verifier that demands held-out transfer and no regression—nothing more exotic—reduced shipped fakes to zero in our experiments while still adopting real improvements; its statistical form both accepted a real embodied repair and rejected the same candidate at deployment scope before it could ship catastrophic forgetting; and under a ten-recipe search it held the capability floor while honestly reporting that no recipe earned adoption. If self-improving agents are to be trusted with their own modification stream, the verifier is not a component of the loop. **The verifier is the loop; the rest is a proposal generator.**

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A Reproducibility

All numbers in this paper are produced by scripts in the accompanying repository (anonymized for review); each result entry in the repository’s ledger names the script and exact configuration that produced it. Text and code experiments run on a single GPU with a 7B open-weight model; embodied experiments use CARLA 0.9.15 + Bench2Drive with the LEAD stack on a single RTX 5090 (Windows-native). The gate itself is released as a dependency-free library (`vsi_gate.py`) exposing `gate()` (score form) and `gate_rate()` (exact-binomial failure-rate form), with a deterministic no-LLM demo.

B Pre-registered decision rules (embodied)

Fixed before the corresponding runs, unchanged throughout: **(local panel)** fix arm: route 11755, 8 fresh rollouts, pass iff ≤ 2 failures; retention arm: the 12 confirmed-regression routes $\times 2$, pass iff pooled failures $\leq 30\%$; both required. **(deployment gate)** screen all 169 baseline-clean routes once; flag any failure; triage pure-artifact flags (minimum-speed/lane-only) out; confirm each flagged route with interleaved candidate/baseline runs (4+4); confirmed regression iff candidate $\geq 3/4$ failures and baseline $\leq 1/4$; global REJECT iff ≥ 1 confirmed regression or pooled candidate failure significantly above baseline (one-sided exact test, $\alpha = 0.05$). **(rate gate)** a fix claim requires a one-sided exact-binomial significant reduction against the measured baseline rate, power-planned before collection (e.g. $58\% \rightarrow 20\%$ requires ~ 19 runs/arm at 80% power). These rules are reproduced verbatim as pre-registered; the main text reports the more conservative two-sample Fisher’s exact p -values computed from the same raw counts—every verdict is unchanged under both tests.